

POST-DEPLOYMENT MEDICAL ASSESSMENT THE POST-TROPICAL SCREENING

Version 2004

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1. Introduction

The term post-deployment medical assessment in general relates to ‘check-ups’ of the physical and mental condition of personnel returning after deployment. This lecture will focus on assessments regarding infectious diseases, often referred to as ‘*post-tropical screening*’. It is assumed that personnel with clear signs or symptoms will undergo (or have already undergone) specific diagnostic assessment. The post-tropical screening is therefore mainly concerned with *latent, symptomless* infections, although in practice minor symptomatic illnesses may be discovered that have not yet been presented at other occasions.

There is a general tendency in post-tropical check-ups to test for all sorts of latent infections without prior consideration of pathogenicity, treatment consequences and cost-effectiveness of the diagnostic strategy. It is however important to note that many latent (mainly parasitic) infections are self-limiting without causing any harm, and that established diagnostic strategies for clinical infections may not be the most (cost-)effective to discover latent cases. For example, a large-scale prospective study from Canada showed eosinophilia, which is often used as a marker for worm infections, to have a low predictive value for latent schistosomiasis, filariasis or strongyloidiasis. Increasingly, the primary aim of post-tropical screening is regarded as *excluding* the presence of latent infections that may cause short- or long-term damage to the health of the patient or his human environment. It should therefore have clearly stated objectives in terms of infections screened for, diagnostic procedures and treatment. These objectives should in turn be based on exposure risks, i.e. geography and activities during deployment, cost-effectiveness (e.g. as costs per long-term case of illness averted) and availability of safe and effective treatment.

2. General aspects of the screening procedure

The basic post-tropical check-up involves medical history, physical examination and general laboratory investigations. Further diagnostics can be done when indicated.

2.1. medical history

Apart from a general organ-system based history, special attention should be paid to symptoms of and risk factors for infections prevalent in the area of deployment. Since geography and activities are generally known, a specific checklist can be made in advance. Examples are fresh water exposure and urinary symptoms in areas that are endemic for *Schistosoma haematobium*, and exposure to tsetse flies and a history of a trypanosome chancre in areas with sleeping sickness. Important symptoms to ask about are gastrointestinal impairment, pruritus and other skin abnormalities, and indicators of chronic infection such as weight loss, low-grade fever and night sweats. All patients should be asked about unprotected sexual intercourse and symptoms of STDs.

2.2. physical examination

If the history is unremarkable, physical examination will rarely reveal abnormalities. In such patients, a general examination with special attention to skin, lymph glands, liver, spleen and genitals will usually suffice. The examination can be extended as guided by symptoms present.

2.3. laboratory investigations

In case of a negative history and physical exam, the laboratory investigations can generally be restricted to:

- general haematology, including ESR, haemoglobin, haematocrit, leukocyte count and differentiation. The most frequent abnormality is eosinophilia. Blood chemistry (e.g. liver enzymes, BUN, glucose) is generally negative in non-symptomatic patients.
- simple urinalysis, including glucose, albumin and sediment microscopy. Microscopic haematuria is an important sign of urinary schistosomiasis.
- stool parasitology: microscopic stool examination for worm eggs and parasitic cysts using a concentration method (e.g. Ridley) is often positive, although the majority of cysts are non-pathogenic. A single specimen test is usually too insensitive to exclude latent infections with pathogens as *Strongyloides stercoralis*, *Schistosoma mansoni* or *Giardia lamblia*; the required number of specimens should be based on the screening objectives (see ‘screening strategies’ below). Vegetative parasitic stages are rarely present without symptoms. In non-symptomatic patients the time consuming examination of direct (i.e. unstained) specimens is generally not required.

2.4. additional diagnostics

Additional investigations may be warranted by abnormalities found. Strictly speaking, this is no longer the stage of screening, but of diagnostic assessment. More often, additional studies are done as part of a specific screening strategy. Most of these will involve serology. In non-symptomatic patients with no abnormalities on physical examination, chest X-rays will rarely provide relevant information.

2.45. time of assessment

Many infections eligible for screening (e.g. schistosomiasis, tuberculosis, intestinal helminthiasis) have incubation or prepatent periods of several weeks, i.e. laboratory tests will remain negative until this period has passed. Unless symptoms are present, the check-up can therefore best be postponed until 6 to 8 weeks after return. Such policy may however interfere with logistical requirements and does not allow timely assessment in case of symptoms. A practical solution may be a two-step approach of a quick screening on return (e.g. questionnaire-based) followed by a more extensive screening including laboratory tests after 6 to 8 weeks.

3. Screening strategies

This paragraph suggests diagnostic strategies for some important infections eligible for screening. Cost-effectiveness considerations are taken into account, but are subject to availability of specific diagnostic facilities. Since exclusion of latent infection is the primary objective, high sensitivity, at least in the first step, should be the most important feature of the diagnostic method of choice. If necessary, more specific and less sensitive tests can then be used as a secondary, confirmatory step.

3.1. tuberculosis

In non-symptomatic patients, screening for infection with *Mycobacterium tuberculosis* may be useful to initiate follow-up, but especially when chemoprophylaxis is considered. Chemoprophylaxis, or treatment of latent tuberculosis infection (LTBI), is aimed at preventing progression of latent infection to active tuberculosis (also known as TB disease). Screening of LTBI can be done by a tuberculin skin test (TST) performed 6-8 weeks after return. In case of frequent missions, screening can be done at 6 to 12 month intervals. Chest X-rays have limited use for tuberculosis screening unless the medical history cannot be relied upon.

A positive TST reflects the presence of delayed-type hypersensitivity (DTH) to mycobacterial antigens *in general*. The interpretation of a TST reaction depends on the sensitivity and the specificity of the test at a given cut-off level. False-negative reactions may occur in case of cellular immune suppression ('anergy') and in the elderly. False-positive reactions can be caused by prior BCG-immunisation, boosting and cross-reactivity to non-tuberculosis ('non-human' or 'atypical') *mycobacteria*. BCG has little effect on the TST reaction if vaccination was only given at birth (as in many countries), but may then still cause boosting (see below). Since the prevalence of atypical mycobacterial infections differs between geographic regions (e.g. higher in tropical climates), the specificity of the TST varies between populations. Most countries restrict treatment of LTBI to persons who were recently infected (generally defined as within the previous 2 years) because these have the highest risk of progression to TB disease. The probability that a positive TST reflects recent infection depends on the background prevalence of 'old' TB latent infections. This will differ between populations and, within populations, between age groups (generally higher with older age). In addition, it will depend on the probability that recent infection indeed occurred. This 'pre-test probability' will be low in many cases but high in some (e.g. close contacts of infectious TB patients, health care workers in countries with high TB incidence). In low-incidence countries such as the Netherlands, the positive predictive value for recent LTBI of a single TST reaction of 10 mm will thereby be low unless such high pre-test probability exists. This is important because a low positive predictive value implies many false-positive cases and thereby many unnecessary LTBI treatments. Guidelines for tuberculin skin testing in low-incidence countries therefore increasingly recommend different cut-off values for different situations. For example, guidelines in the USA and The Netherlands recommend 10 mm as the cut-off for close contacts of TB patients, but 15 mm if there is clear history of TB exposure.

To avoid these complications in interpretation of a single TST result it is recommended that for military deployment to high-incidence countries a baseline TST reaction is determined before departure. Subsequent increase in TST reaction size above a set limit ('TST conversion') then indicates TB infection during the mission. However, the possibility of a booster response should be taken into account. A booster response (or boosting) is a false-positive TST conversion between two tests due to recall of waned DTH by the tuberculin antigens of the first test in a patient who has been previously sensitised to mycobacterial antigens. In elderly patients, this may reflect tuberculosis infection during childhood; in younger patients it is often due to BCG immunisation in the past. Boosting may occur if tuberculin tests are given within a short time span (maximum 1-6 weeks, but may occur up to 12 months), such as prior to and after deployment. As it will only occur following the first test, it can be excluded by repeating the pre-deployment test after two weeks and recording the second result as the baseline value ("initial 2-step testing"). This is particularly important for personnel with a history of BCG vaccination but a negative baseline test. If no baseline test result is available, the positive predictive value of a single TST reaction (i.e. the probability that a positive reaction is due to LTBI) depends not only on the test specificity

Whether or not a baseline test is done, cut-off values for positive TST or conversion for small deployments are best based on national guidelines, for large deployments these can be determined on a case-by-case basis in consultation with specialists in TB control.

Recently developed blood tests may provide a useful alternative to the TST. These tests measure interferon-gamma production (e.g. Quantiferon Gold) or count interferon-gamma producing lymphocytes (e.g. Clinispot) after stimulation with antigens that are highly specific to *M. tuberculosis* and do not cross-react with BCG (and only to limited extent with atypical mycobacteria). These tests have the advantage that they identify latent TB infection with high specificity, also in BCG vaccinated individuals, and do not require a second visit for reading. Studies that evaluate their (cost-)effectiveness for screening in various settings are underway.

3.2. amoebiasis

Non-symptomatic carriage of *Entamoeba histolytica* cysts is a frequent finding in expatriate workers. The majority of cases reflect self-limiting intestinal colonisation by non-invasive, non-pathogenic zymodemes, that have recently been identified as a separate species and named *E. dispar*. Some patients do however carry the potentially invasive *E. histolytica* sensu stricto, that is capable of causing extra-intestinal pathology (such as liver abscesses) even after long latency periods. Amoebic cysts are directly transmissible between humans.

Passage of *E. histolytica* cysts has generally been regarded as an indication for treatment. Several techniques are available for differentiation between *E. histolytica* and *E. dispar*, including polymerase chain reaction (PCR) and serologic tests that detect specific antigens in stool specimens. Data are needed to confirm the non-pathogenicity of *E. dispar* and to assess the cost-effectiveness of routine species identification versus treatment of all carriers who will not return to an endemic area at short notice. The role of serum antibody testing in detecting potentially invasive infections remains to be established.

First-line diagnosis of carriage is by microscopic examination of a concentrated stool specimen. The negative predictive value (i.e. its value for *excluding* amoebic carriage) of a single specimen test is difficult to determine, as it depends not only on the accuracy of the laboratory but also on the expected prevalence in the population examined. Most parasitologists regard two specimens as sufficient. Treatment with a contact amoebicide, e.g. diloxanide furoate 500 mg tid for 10 days, is highly effective for eradication of carriage. If unavailable (diloxanide has recently been taken off the market in the Netherlands): clioquinol 250 mg tid for 10 days. Clioquinol is less well tolerated.

Imidazoles such as metronidazole and tinidazole are generally first choice tissue amoebicides but less effective for eradication of carriage.

3.3. schistosomiasis

Latent or symptom-poor infections with *Schistosoma mansoni*, *S. haematobium* and, to a lesser extent, *S. intercalatum*, are frequent findings following fresh water exposure in endemic areas. Infections with *S. japonicum* are rarely encountered outside endemic populations, but may be acquired during missions in Southeast Asia. *S. japonicum* generally causes (severe) symptomatic disease and is therefore less relevant for screening.

The clinical presentation of schistosomiasis in travellers and expatriates differs from that in endemic populations. Apart from acute allergic reactions to egg antigens (Katayama syndrome), patients may present with abdominal symptoms and diarrhoea (*S. mansoni*, *S. intercalatum*), urinary symptoms and microscopic haematuria (*S. haematobium*), as well as ectopic granulomas and general symptoms such as chronic fatigue. Irreversible portal or renal pathology, which are characteristic of chronic schistosomiasis in endemic populations, are extremely rare, possibly due to lower worm loads and lack of exposure to egg antigens in early life or even in utero. Subclinical infections in travellers may however result in nerve system disorders due to granulomatous lesions around disseminated schistosoma eggs in the spinal cord or supplying arterioles

(neuroschistosomiasis). Such complications are increasingly reported following exposure in sub-Saharan Africa (notably Lake Malawi) to *S. mansoni* or *S. haematobium*, and may occur without Katayama syndrome or other symptoms until 6 or more months after the exposure. Both latent and clinical schistosomiasis may occur in the absence of eosinophilia.

Secondary prevention of neuroschistosomiasis is a major reason for screening. In latent infections, serology has much higher sensitivity than microscopic examination of stools or urine. Many tests have been developed which use various egg- and somatic antigens. As both short and long-term pathology are caused by immunologic reactions to schistosomal eggs, a test based on egg antigens (usually an ELISA) is generally regarded as the most cost-effective first step. Seroconversion can be demonstrated after 6-10 weeks following exposure. Tests (e.g. immunofluorescence) that use somatic antigens, which may become positive earlier, i.e. before egg production starts, have little additional value for screening purposes, because praziquantel treatment is less effective in this stage but may also elicit a Katayama reaction. In addition, non-productive infections (rather frequent in minor exposures) can safely remain untreated. However, in a small percentage of *Schistosoma* infections egg production may not start until several months after infection.

If antibodies to egg antigens are demonstrated, the infection may be confirmed by examination of stool, urine or biopsies of the rectal mucosa ('rectal snips'). Although not required for treatment, it may provide a baseline value for follow-up, as antibody titres may remain elevated for several months (probably years) after treatment, and treatment may be incomplete. Absence of eosinophilia has limited negative predictive value for latent *Schistosoma* infection.

Treatment with praziquantel, 40 mg/kg bodyweight in a single dosis, eradicates most infections although some viable worms may remain. Treatment for three days has been advocated for complete eradication, but there is little evidence that this is more effective than single dose treatment.

3.4. strongyloidiasis

Symptomatic infections with *Strongyloides stercoralis* after return from endemic areas are rare. Abdominal complaints, migratory skin lesions (larva currens) and weight loss are the main symptoms. Most infections remain non-symptomatic. As the laboratory diagnosis is difficult, particularly in low-grade infections, little is known about the prevalence of such latent infections.

The infection may persist throughout life due to an auto-infective cycle, and can result in a life threatening hyperinfection syndrome involving paralytic ileus and gram-negative septicaemia when the immunity is suppressed (e.g. by high-dose steroid therapy) at later age. High prevalences of latent infections have been found after more than 30 years in British and Dutch veterans who were prisoner-of-war in Southeast Asia during WW II. Transmission occurs through bare skin exposure to moist, warm soil, usually by walking barefoot.

Symptomatic strongyloidiasis is generally characterised by hypereosinophilia, but eosinophil counts are normal in up to one third of latent infections. Microscopic examination of stained or unstained stool specimens concentrated by routine methods (e.g. Ridley) has low sensitivity. Stool culture and examination of duodenal aspirates (e.g. by the 'string test' method) are more sensitive but quite elaborate, and are therefore of limited practical use for screening purposes. A comparative study of several diagnostic methods in Dutch WW II veterans showed that examination of 3 stool specimens concentrated by the Baermann method was the most sensitive strategy. As in all microscopic methods, this may depend on the laboratory.

Serologic methods are increasingly used for screening purposes and probably the method of choice for the first screening step. Most used are tests based on filarial antigens with which antibodies to *S. stercoralis* often cross-react. Intermediate titres in the absence of homologous infection are then considered suggestive of *Strongyloides* infection. Data on sensitivity are however limited, and specificity may be a major problem, as many cases will need confirmation. Tests based on non-human *Strongyloides* species such as *S. ratti* are reported to have high (85-90%) sensitivity, but are

probably less specific than those based on homologous *S. stercoralis* antigen. A disadvantage of tests based on *S. stercoralis* may be the relatively high cost, as the antigens must be prepared from stool cultures derived from human specimens. Confirmation of positive serologic results by examination of 3 stool specimens using the Baermann method is generally recommended.

Treatment with albendazol, 400 mg for 3 days, eradicates a large but unknown proportion of non-symptomatic and symptomatic infections in immunocompetent patients. Ivermectin, 200 microgram per kg body weight in a single dose, is probably more effective; this drug is however not licensed or difficult to obtain in many countries.

3.5. other infections

giardiasis

Infections with *Giardia lamblia* are frequent and often symptomatic. The combination of mild diarrhoea, sometimes with steatorrhoea, and upper abdominal symptoms including stomach pains, ructus and loss of appetite, is highly suggestive. Although giardiasis is self-limiting, latent infections may persist for prolonged periods and cause symptoms intermittently. In addition, it is easily transmitted to family members. Screening of asymptomatic patients may have limited cost-effectiveness, as the sensitivity of stool examination is low (mainly due to intermittent excretion of cysts). When clinical suspicion is high, empirical treatment, e.g. single dose tinidazol 50 mg/kg body weight up to 2000 mg, may be considered.

non-pathogenic intestinal parasites

Cysts of non-pathogenic amoebae, such as *Entamoeba coli*, *Entamoeba hartmanni*, *Endolimax nana* and *Iodamoeba buetschlii* are also frequently found. Treatment is not indicated. Non-pathogenic flagellates, such as *Chilomastix mesnili* do not require treatment either. The pathogenic potential of *Blastocystis hominis* is still matter of debate. Treatment should only be considered in symptomatic cases (diarrhoea) if several stool examinations did not reveal other causes.

filariasis

The most important filariases that cause symptoms in humans (lymphatic filariasis, onchocerciasis or river blindness, and loiasis) are rather frequently encountered in expatriate workers and troops that have been highly exposed to vectors in endemic areas. Screening of latent infections is generally of little use, since all filariases have long (6 months to 17 years) prepatent periods during which diagnostic tests are negative; only a minority of such infections result in clinical disease; chronic irreversible damage is extremely rare in non-endemic populations; and asymptomatic infections are harmless and do not need treatment. Treatment of latent infections may even cause severe allergic reactions due to sudden release of egg antigens. Personnel returned from endemic areas should be informed about symptoms that may occur later and be reassured about the reversible nature of symptomatic infections.

sexually transmitted diseases

Gonorrhoea, syphilis and Chlamydia infection may be present without symptoms and have health consequences in the long run. These infections will be addressed in detail in the lecture on STDs.

4. Conclusion

Post-tropical assessment of military personnel can be useful. It may involve a specific medical history, physical examination and limited laboratory investigations. Specific screening for non-symptomatic infections is increasingly advocated, but should be limited to those infections that may cause harm in the long run and can be adequately treated. There should be clear screening policies based on cost-effective diagnostic strategies.